

Data Exploration for Your Final Project

Finding a Dataset for Target Trial Emulation

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Welcome to the data exploration hub for your final project. Finding the right dataset is arguably the most important step in a successful causal inference study. A good dataset doesn't just have interesting variables; it must have the right **structure** to allow for the emulation of a target trial.

This page provides you with two things: a powerful tool for searching broadly and a curated list of datasets that are known to be well-suited for this kind of work.

The Go-To Tool: Google Dataset Search

Think of this as Google Scholar, but for data. The **Google Dataset Search** engine indexes thousands of data repositories from around the world. It is the best place to start when you have a specific topic in mind.

Launch Google Dataset Search

Tips for Effective Searching

When searching, use keywords that describe the **structure** of the data you need, not just the topic. Good causal inference requires data where time is well-defined.

- Combine your topic (e.g., "cancer", "education", "air pollution") with structural keywords like:
- "longitudinal study"
- "cohort study"
- "prospective data"
- "panel data"
- "linked mortality data"

Curated Datasets for Target Trial Emulation

Not all datasets are created equal. The best datasets for TTE are **longitudinal cohort studies**, where a group of people is enrolled at a baseline point in time and followed forward. This structure allows us to properly define eligibility, measure confounders before treatment, and follow individuals for the outcome.

Here are a few high-quality, publicly-available datasets that are exceptionally well-suited for your project.

Framingham Heart Study

- **Why it's great for TTE:** This is the archetypal prospective cohort study. A cohort of residents from Framingham, MA, was enrolled in 1948 and has been followed ever since. It contains detailed baseline measurements of potential confounders (e.g., smoking, cholesterol, blood pressure) and tracks cardiovascular outcomes over many decades. It's a perfect dataset for emulating a trial of an intervention (e.g., lifestyle change, medication) on heart disease.
- **Access:** Data is available to researchers through a proposal process. [Visit the Framingham Heart Study Website.](#)

MIMIC-IV (Medical Information Mart for Intensive Care)

- **Why it's great for TTE:** This database of ICU admissions contains high-resolution, time-stamped data. This is critical because you can precisely define **time zero** (e.g., time of admission, time of a specific diagnosis). Treatments (medications, ventilators) and outcomes are all logged with precise timing, allowing you to emulate a trial of a clinical intervention in a critical care setting and avoid biases like immortal time bias.
- **Access:** Requires completion of a human subjects research training course. [Visit the MIMIC-IV Website.](#)

Add Health (The National Longitudinal Study of Adolescent to Adult Health)

- **Why it's great for TTE:** This is a classic longitudinal cohort. A representative sample of U.S. adolescents was enrolled in 1994-95 and has been followed into adulthood. This is ideal for questions where the "treatment" occurs in adolescence (e.g., participation in sports, a specific educational program) and the outcome occurs in adulthood (e.g., income, health status). The baseline wave contains rich data on potential confounders (family background, school environment, etc.).
- **Access:** Public-use data is readily available. [Visit the Add Health Website.](#)

NHANES (National Health and Nutrition Examination Survey) with Linked Mortality Data

- **Why it's great for TTE:** By itself, NHANES is a series of cross-sectional surveys. However, participants can be linked to the National Death Index. This transforms the dataset into a prospective cohort study. You can use the rich survey data at baseline (diet, lab values, lifestyle) to define eligibility and measure confounders. You can then emulate a trial where the outcome is long-term mortality.
 - **Access:** Data is publicly available from the CDC. [Visit the NHANES Website.](#)
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Checklist for Evaluating a Dataset

When you find a dataset that looks promising, use this checklist to assess its suitability for a Target Trial Emulation project.

- ☐ **Can you define a clear time zero?** Is there a variable that marks the start of follow-up for all individuals (e.g., date of enrollment, date of diagnosis, start of the school year)?
- ☐ **Are confounders measured at or before time zero?** The fundamental requirement for confounding control is that your covariates are measured *before* the treatment is assigned.
- ☐ **Is the exposure/treatment variable well-defined?** Can you clearly determine who is “treated” and “untreated” based on the data?
- ☐ **Is the outcome measured after time zero?** The dataset must have follow-up data to ascertain the outcome after the intervention.
- ☐ **Is the follow-up period long enough?** Is there sufficient time between treatment and outcome for a causal effect to plausibly occur?

If you can answer “yes” to these questions, you likely have a strong candidate dataset for your project.